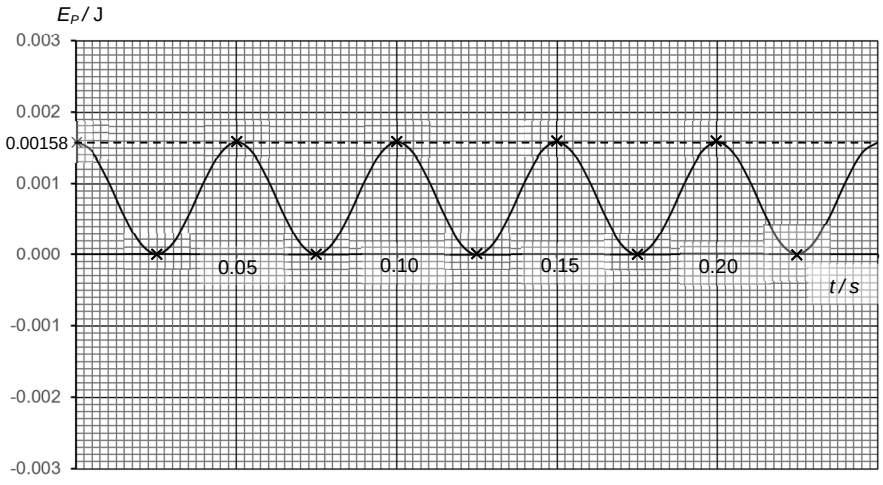
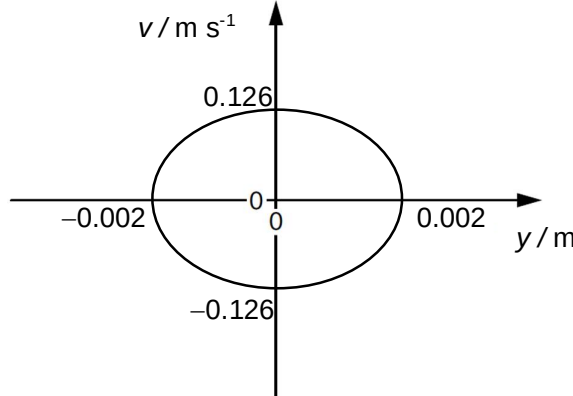


Qns	Answer	Marks
3(a)(i)	Free oscillations are oscillations with constant amplitude and <u>without</u> energy loss or gain as there is no driving or resistive forces acting on it.	B1
3(a)(ii)	Natural frequency is the frequency at which a body will vibrate when there is <u>no driving</u> or <u>no resistive forces</u> acting on it.	B1
3(b)(i)	 $E_{p(max)} = \frac{1}{2} m y^2 \omega^2 = \frac{1}{2} (0.200) (0.002)^2 \left(\frac{2\pi}{0.1} \right)^2$ $= 0.00158 \text{ J}$	B1 correct sine-square shape, all above x-axis B1 correct period & phase B1 correct max E_p indicated
3(b)(ii)	 $v_o = \omega y_o = \frac{2\pi}{T} y_o = \frac{2\pi}{0.1} (0.002) = 0.126 \text{ m s}^{-1}$	B1 Correct & smooth shape B1 correct max v calcn at $y=0$ B1 correct max y at $v=0$
3(b)(iii)	Period decrease. Equilibrium point is raised once the mass is removed. Hence amplitude increases.	B1 B1

Qns	Answer	Marks
4 (a)(i)	The Principle of Conservation of Linear Momentum states that	D1